

Emma Lucia Byrnes Finn

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Education

Oxford University

Incoming DPhil student in Statistics supervised by Prof. Yee Whye Teh and Prof. Patrick Rebeschini

Oxford, UK

Matriculating October 2026

Harvard University

A.B, Mathematics and Classics, *magna cum laude* with Highest Honors in Mathematics. GPA 3.97/4.00

Cambridge, MA

May 2026

Concurrent A.M, Statistics, GPA 3.92 /4.00.

May 2026

Honors Thesis in Mathematics: *From Here to There: Transport Entropy Inequalities*, (advisors: Dr. Mark Sellke and Dr. Kevin Yang)

Thesis in Classics: *Quantifying the Past: Empirical Tropes in Greek Historiography*, (advisor: Dr. Emily Greenwood)

Honors and Awards

Rhodes Scholarship

November 2025 – Present

Rhodes Trust, awarded fully funded postgraduate study at the University of Oxford

Concurrent Masters Prize in Statistics

May 2026

Award given by the Harvard Statistics Department recognizing outstanding work in the concurrent AM program

David Mumford Undergraduate Mathematics Prize

May 2026

Award given annually to the most promising senior concentrator in the Harvard mathematics department

Thomas Temple Hoopes Prize

May 2026

Harvard-wide award for outstanding undergraduate scholarly work on the basis of my Classics thesis

Classics Department History Thesis Prize

May 2026

Harvard Classics department prize, voted by the faculty, for the best undergraduate thesis on a historical subject

Marshall Scholarship, National Finalist

November 2025

Marshall Aid Commemoration Commission, selected as a finalist for a UK postgraduate scholarship

Phi Beta Kappa Inductee

November 2025

One of 48 Harvard Seniors inducted to the United States' oldest academic honor society recognizing exceptional academic achievement

Coolidge Scholarship

June 2021 – May 2026

Calvin Coolidge Presidential Foundation, awarded a four-year, full-ride merit-based scholarship to any US college

Best Poster Award, Kranium Summer Research Program

August 2024

Kempner Institute at Harvard University

Kempner Institute Research Fellowships

May 2024 – May 2025

Kempner Institute at Harvard University, selected for Summer 2024, Fall 2024, and Spring 2025 Kempner Institute research cohort

John Harvard Scholar

September 2023 – September 2024

Harvard University, awarded to the top 5% of the class by GPA

Selected Publications and Preprints

Wang, B., Finn, E., and Liu, B. *The Two Clocks and the Innovation Window: When and How Generative Models Learn Rules*. NeurIPS 2026, under review.

Finn, E., Wang, B., Keller, T. A., Ba, D.E. *A Wavelet View of Diffusion*. AISTATS 2026, accepted.

Finn, E. *“Not by Reason Alone:” Plato’s Phaedrus and Fictionality*. Alexandria: The Oxford Undergraduate Classics Journal, accepted.

Wang, B., Finn, E., and Liu, B. *When Rule Learning Breaks: Diffusion Fails to Learn Parity of Many Bits*. NeurIPS 2025, Structured Probabilistic Inference and Generative Modeling: Probabilistic Inference in the Era of Large Foundation Models, accepted (**oral presentation**).

Finn, E., Wang, B., Keller T. A., Ba, D.E. *Where the Score Lives: A Wavelet View of Diffusion*. NeurIPS 2025, Structured Probabilistic Inference and Generative Modeling: Probabilistic Inference in the Era of Large Foundation Models, accepted.

Finn, E., Keller T. A., Theodosis, M., and Ba, D.E. *Origins of Creativity in Attention-Based Diffusion Models*. 3rd ICML Workshop on High-dimensional Learning Dynamics, accepted.

Finn, E., Keller T. A., Theodosis, M., and Ba, D.E. *Learning Artistic Signatures: Symmetry Discovery and Style Transfer*. arXiv:2412.04441, 2024, preprint.

Invited Talks and Presentations

Finn, E. *Where The Score Lives: A Wavelet View of Diffusion*, Microsoft Research New England Generative Modeling and Sampling Seminar, Microsoft Research, Cambridge, MA, April 2026.

Expository Work

Finn, E. *Score and Structure*. Author of an ongoing blog that explores the connections between algebra, probability, and ML, while making advanced theory accessible. emmafinn314.github.io/blog.

Research, Teaching, and Professional Experience

CRISP Lab

Undergraduate Researcher

Cambridge, MA
December 2023 – Present

- Analyzed how properties of the data distribution affect score learning in diffusion models by proposing an analytically tractable parametrization of the score function and characterizing optimal solutions under this model
- Demonstrated a concrete failure case for learning parity rules in diffusion models and proposed a benchmark to evaluate memorization, generalization, and rule-learning accuracy
- Studied generalization and inductive bias in attention-based diffusion models; designed custom datasets to probe which classes of rules such architectures can learn

Harvard Statistics Department

Teaching Fellow

Cambridge, MA
September 2024 – Present

- Courses: STAT 110 (Intro to Probability, Fall '24), STAT 111 (Intro to Statistical Inference, Spring '25 and Spring '26) and STAT 210 (Graduate Probability I, Fall '25)
- Taught weekly sections; wrote original review content, homework solutions, and practice exam solution guides; graded homework and exams
- Hosted three review sessions (200+ students total); achieved score of 4.95/5 across student evaluations for effectiveness and engagement

Harvard Math Department

Course Assistant

Cambridge, MA
August 2023 – May 2024

- Facilitated teaching of multivariable calculus (Fall 2023) and linear algebra (Spring 2024) to undergraduate students
- Led problem-solving workshops to illustrate applications of course material, held office hours, and graded homework

Federal Reserve Bank of St. Louis

Government Relations Intern

St. Louis, MO
May – August 2023

- Directed a project to analyze data from 300+ meetings with federal and state officials over four years to identify underrepresented constituencies in the Eighth District of the Federal Reserve and support equitable outreach

Service and Leadership

Statisticians Without Borders

Volunteer Statistician

Remote
July 2025 – Present

- Selected to join the pro-bono arm of the American Statistical Association, consulting for Scholars at Risk on databases to defend academic freedom

Kempner Institute Editorial Board

Reviewer and Editor

Cambridge, MA
July 2025 – Present

- Provided scientific review for the Kempner's public communications, ensuring published articles were technically sound

Kempner Institute Lunch and Learn Series

Organizer and Presenter

Cambridge, MA
August 2024 – Present

- Coordinated weekly forum for researchers to discuss machine learning and neuroscience; led outreach to undergraduates, fostered inclusive dialogue across disciplines, and helped speakers to make their work accessible; delivered three talks
- Doubled overall attendance through outreach and tripled the number of undergraduates who presented work

Boston Diffusion Day

Founder and Co-organizer

Cambridge, MA
March – October 2025

- Co-founded a research conference with 300+ registrants to build community among Boston diffusion model researchers
- Secured \$5k in funding, developed programming, and led outreach to local academics and industry researchers

Programs and Workshops

Mathematics and Machine Learning Program

Workshop Participant

Cambridge, MA
September – November 2024

- Selected to attend a program at the Center of Mathematical Sciences and Applications at Harvard focused on applications of ML to open mathematical problems

Harvard Math Department

Directed Reading Program Participant

Cambridge, MA
January – December 2024

- Conducted an independent study and wrote an expository paper on stochastic branching processes
- Presented on combinatorial optimization algorithms as part of an independent study of algebraic graph theory

Technical Skills: Python (PyTorch), R, experience running experiments on shared high-performance computing clusters.